MPX · PD-298 · DESIGN REVIEW

Boil-Loop Mini v0.5

Boil, catch the lime-crusted plastic, cool, pour

What it does, what the simulation showed, which of our 22 technical requirements it meets and misses, and a **design FMEA**. It removes microplastics well in hard water, but it makes about **1.2 L/h**: far below our 15 L/h target.

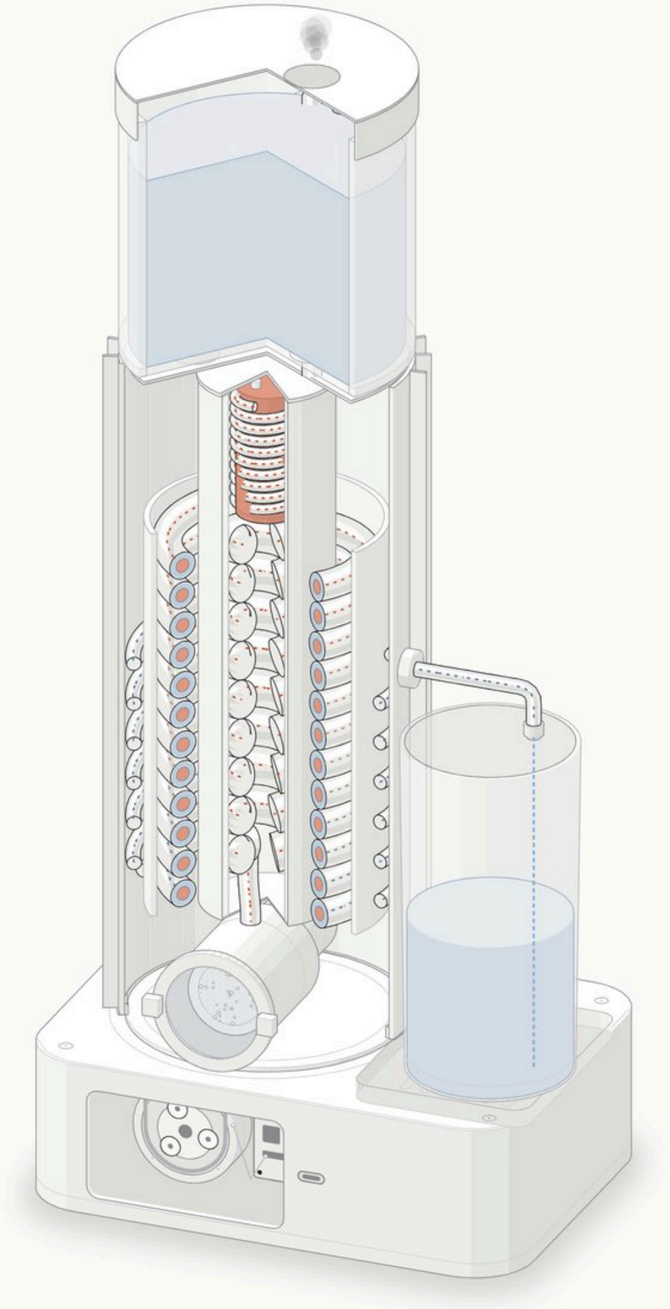
1.2 L/h
vs a 15 L/h target

90 %
removal at 285 mg/L hardness*

2.7 kg
Ø105 × 388 mm, 60 W USB-C

38 min
to first water as drawn*

* simulated (tube model + lumped warm-up run); no physical tests yet



EXPLORATORY CONCEPT · DRAWINGS V0.5 · OCT 2026

Where this concept sits in the CED process

Boil-Loop is the boil-and-filter concept we kept exploring after concept selection. It already has a layout and drawings, so the same embodiment checks apply: **critical review and FMEA**.

LEC 3–5 ✓
Task clarification
Requirements list (22), SNPS, objective tree

LEC 6–8 ✓
Conceptual design
Boil-and-filter came out of the morphological chart

LEC 9 ✓
Simulation
Tube model, chemistry, warm-up run (this deck)

LEC 10 ◑
Evaluation
Not in the chosen three (04, 07, 05): explored as an option

NOW
Embodiment design
v0.5 layout + drawings done; FMEA in this deck

ONLY IF
Detail design
it passes the hardness and flow gates

LATER
Production
Production instructions

EMBODIMENT DESIGN STEPS (LECTURE 2, SLIDE 36) AND WHERE BOIL-LOOP STANDS

✓ Overall layout: general arrangement, spatial compatibility	v0.5 drawing set S01–S08, STEP assembly, Ø105 × 388 mm
✓ Preliminary form design: shapes, materials, processes	316L tube coils, Al finned shell, borosilicate jug, 7 electrical parts, no PCB
◑ Apply all major criteria	Technical by simulation. Flow, mass (2.7 kg) and cost fail or are open
◑ Develop with scale drawings and critical reviews	Drawings done; this deck is the critical review
▶ **Use methods for identification of errors (e.g. FMEA)**	Draft design FMEA: 20 failure modes, 11 critical (slides 13–16)
○ Apply methods for optimisation	Only worth it if the two gates pass: hardness of our water, and a flow the team accepts
○ Outcome: embodiment ready for detail design	Not yet

TWO GATES BEFORE MORE EMBODIMENT WORK

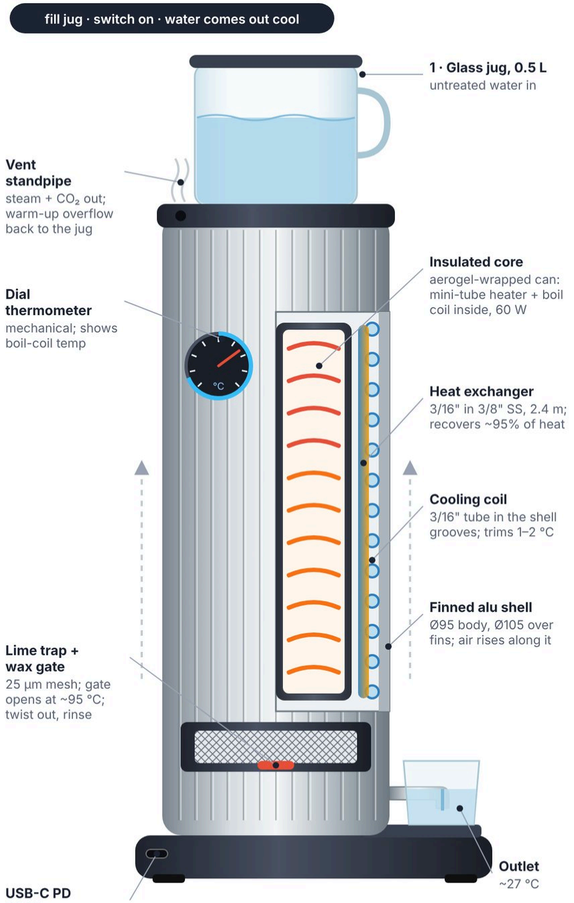
The FMEA is useful now, but two findings decide whether the concept is worth detailing at all.

- **Gate 1, chemistry:** measure the hardness of the water we target. Removal is 90 % at 285 mg/L but only about 40 % at 80 mg/L.
- **Gate 2, flow:** 1.2 L/h is fixed by the 3 min boil and the 60 W heater. 15 L/h would need about **580 W** and a **750 mL** boil coil.
- If both pass, carry on with the FMEA actions; if not, keep it as a bench experiment, not a product.

Sources: CED lec02 sl. 36–37; Pahl & Beitz, Engineering Design 3e, §10.4; MPx project notes

What the product does

A bottle-sized purifier on USB-C. It pumps water through thin steel tubes, holds it at a boil so dissolved lime crusts around microplastics, catches the crust while hot, then cools the water with its own heat exchanger.



1 Fill and switch on

Pour 0.5 L into the glass jug and plug in a 65 W USB-C charger or power bank. A peristaltic pump pulls **20 mL/min** from the jug.

2 Heat with its own heat

Cold water climbs the outer tube of a 2.4 m tube-in-tube exchanger and is warmed to about **95 °C** by the water coming back down. A **60 W** heater adds the last few degrees.

3 Boil and crust

About **3 min** at the boil in a vented 61 mL coil. Calcium bicarbonate loses CO_2 and comes out as lime, which crusts around plastic particles.

4 Catch and cool

A **25 µm** steel mesh in a twist-out drawer catches the crust. A wax gate only opens above 92–95 °C. The exchanger and the body coil cool the water to about **27 °C**.

KEY NUMBERS

Size	Ø105 × 388 mm (over fins and jug)
Mass	2.73 kg dry. The concept sheet said 1.3 kg; the exchanger and shell dominate
Power	USB-C PD 20 V, 60 W heater; about 23 W once running
Flow	1.2 L/h; jug of 0.5 L
Removal	90 % at 285 mg/L hardness; about 40 % at 80 mg/L (Yu et al. 2024 fit)
Water path	316L tube, borosilicate glass, silicone seals; PharMed pump tube on the cold side
Electrics	7 parts, no PCB, battery or sensor; wax gate and standpipe do the control
Cost	Not costed for v0.5. v0.4 BOM: ₹14,890 prototype, ₹6,650 implied retail

AGAINST OUR SNPS

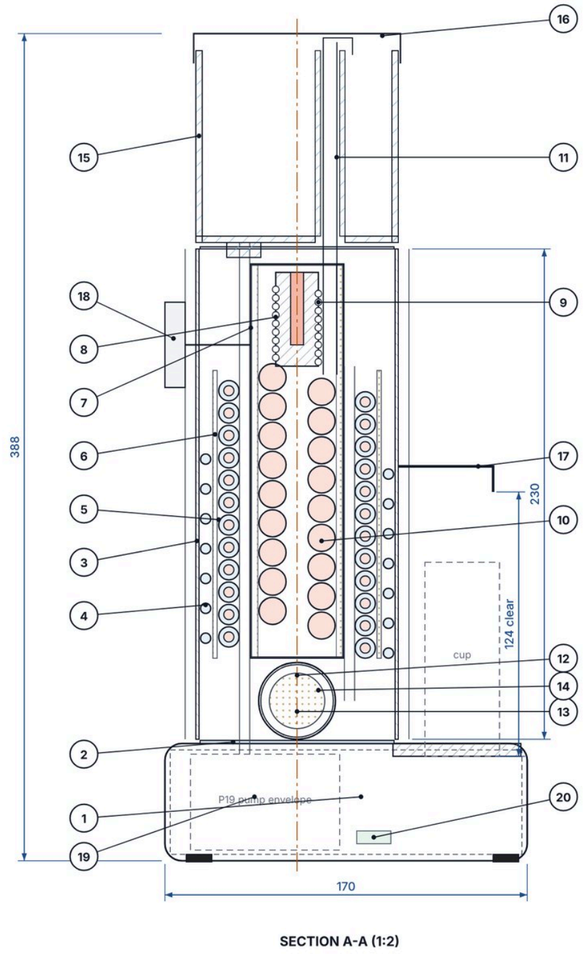
Remove
yes, in hard water

Detect
no; pairs with C05

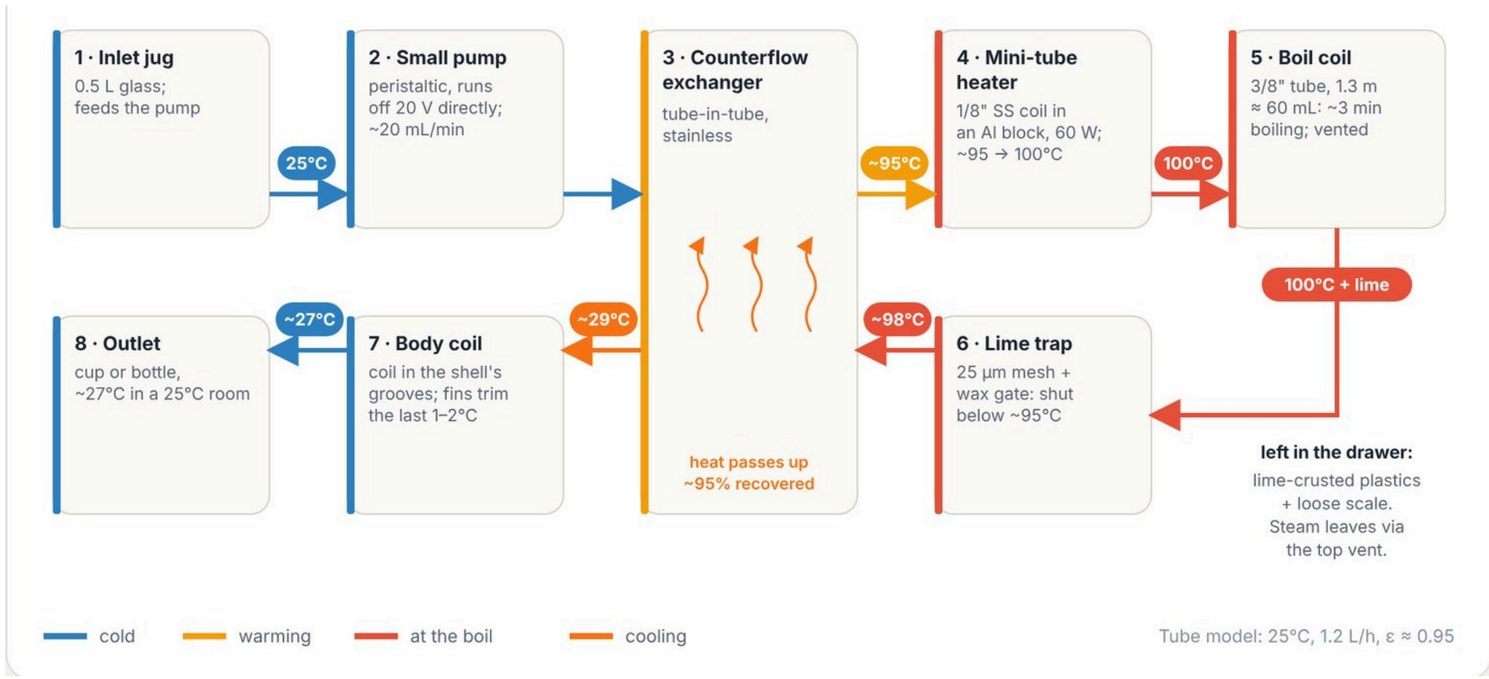
Contain
lime in the drawer

How water moves through it

Top row heats the water, bottom row brings it back down. The exchanger in the middle hands the heat from one row to the other, so the heater only covers the last few degrees.



SECTION A-A (1:2)



Warm-up

Until the core passes 95 °C the wax gate stays shut. Pumped water climbs the vent standpipe and **overflows back into the jug**: no water reaches the spout for the first part of a run.

Control without a board

The wax gate (opens 92–95 °C), the overflow standpipe and two thermal cut-offs on the heater block (130 °C, 152 °C) do all the control.

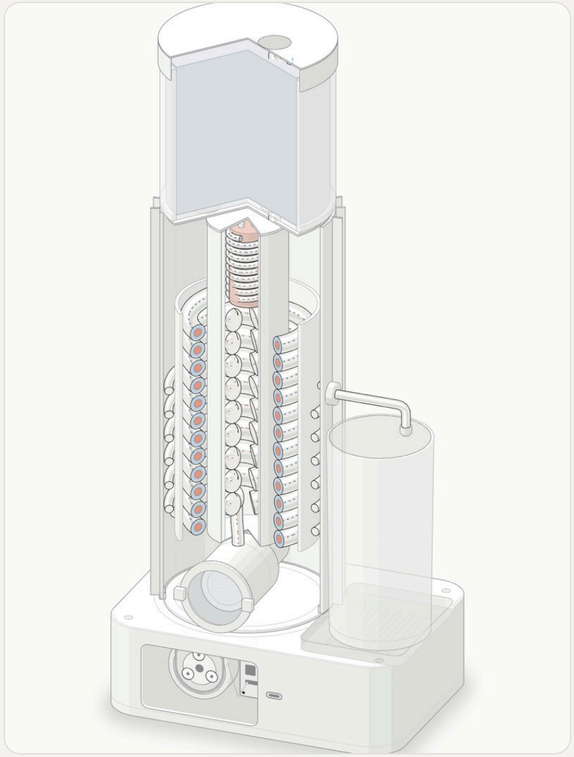
Section A-A

Drawing S01 at 1:2: jug 15 on top, core 7 with heater 8–9 and boil coil 10, exchanger 5, body coil 4 in shell 3, lime drawer 12–14, pump 19 in the base.

Flow diagram from the v0.5 concept sheet (boil coil since changed to 1/2″ × 0.65 m, 61 mL); section from drawing MPx-BLM-S01

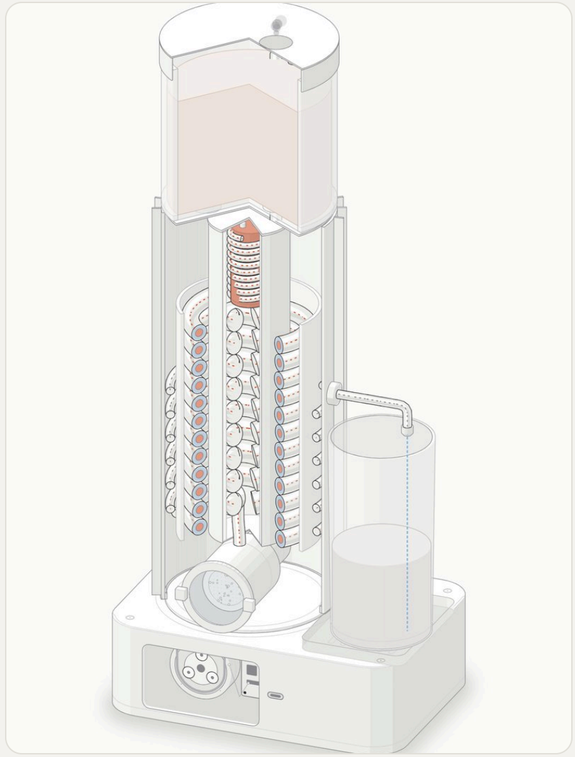
The simulated cutaway: one run from switch-on

An interactive cutaway driven by the same model: tube heat transfer, heater budget, a lumped warm-up run, and the lime chemistry. 1.2 L/h, 285 mg/L, 25 °C room.



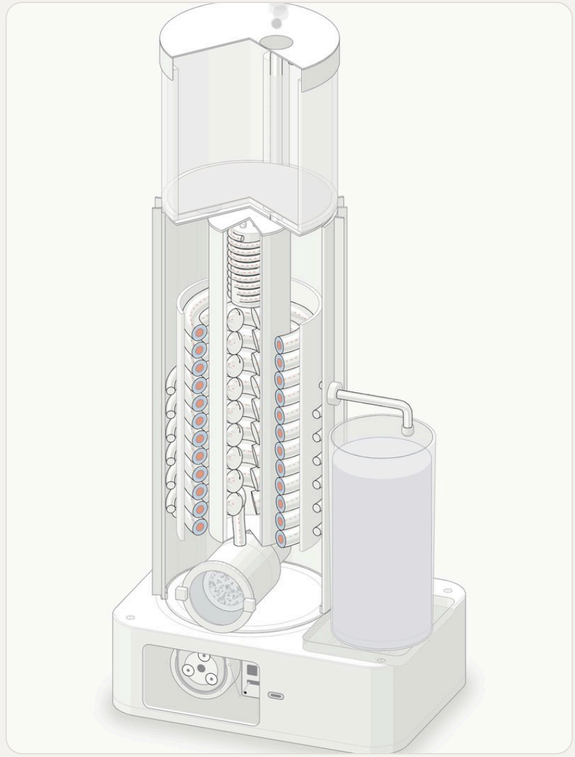
Warm-up, gate shut

Core at 65 °C. The pump runs, but the water overflows back into the jug. Nothing comes out yet.



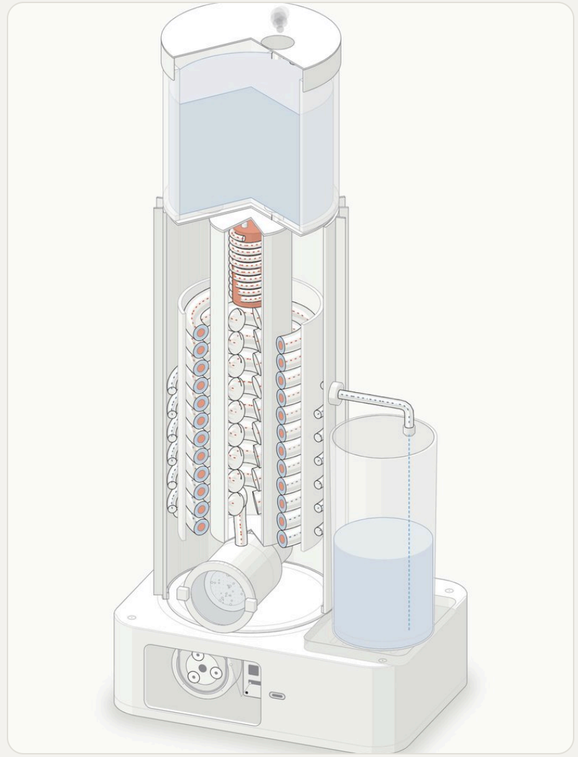
Producing, as drawn

First water at **38 min**. The jug was heated by the overflow, so water leaves at **45 °C**, not 27 °C.



Jug empty

489 mL out after **64 min**, using **65 Wh**.



With a pump thermostat

Proposed fix: the pump waits for a 98 °C core. First water at **9 min**, out at **27 °C**, jug done in 35 min.

Open it: Desktop › MPx › BoilLoop_Mini_v0.5 › figure › MPx_BoilLoop_Mini_v05_Cutaway.html. Sliders for flow, hardness, room temperature and the pump thermostat.

What the cutaway reads out

Temperatures at the eight stations, the steady-state numbers, and what this particular run has done so far.

AS DRAWN, 44 MIN INTO A RUN

01 Jug 58.0°	02 Pump 58.0°	03 Exchanger, cold side 96.6°	04 Mini-tube heater 99.5°
05 Boil coil + vent 99.5°	06 Lime trap + wax gate 99.5°	07 Exchanger hot side + body coil 60.1°	08 Spout 44.1°

Producing. 1.2 L/h at 44.0 °C. The exchanger recovers 95% of the heat, so the 60 W heater runs at about 23 W; the outlet is warm because the jug water was warmed to 57.8 °C during warm-up.

steady at 25 °C room: 22.5 W heater · ε 0.949 · out 27.1 °C · boil 3.0 min · MP removal 90%

this run: t 46 min · first water 38 min · delivered 144 mL · 45.7 Wh

WITH THE PROPOSED PUMP THERMOSTAT, 17 MIN INTO A RUN

01 Jug 25.0°	02 Pump 25.0°	03 Exchanger, cold side 95.2°	04 Mini-tube heater 99.5°
05 Boil coil + vent 99.5°	06 Lime trap + wax gate 99.5°	07 Exchanger hot side + body coil 28.7°	08 Spout 27.0°

Producing. 1.2 L/h at 27.0 °C. The exchanger recovers 95% of the heat, so the 60 W heater runs at about 23 W.

steady at 25 °C room: 22.5 W heater · ε 0.949 · out 27.1 °C · boil 3.0 min · MP removal 90%

this run: t 17 min · first water 9 min · delivered 152 mL · 16.5 Wh

HOW TO READ THEM

- **Steady state looks good:** 22.5 W heater, exchanger effectiveness 0.949, 27.1 °C out, 3.0 min at the boil, 90 % removal.
- **A real run is different:** the warm-up overflow heats the jug to about 58 °C, so the first water arrives at 38 min and leaves at 45 °C.
- **The pump thermostat fixes it:** the jug stays at 25 °C, first water at 9 min, 27 °C out, and 36 Wh per jug instead of 65 Wh.
- **Station 06** reads 99.5 °C: the crust is caught while the water is still at the boil, as the concept intends.

The calculations behind the numbers

Eight hand-checkable equations from the tube model (tube_sizing_sim.py) and the cutaway model (model.mjs).

1 Exchanger effectiveness

$$\varepsilon = (1 - e^{-N(1-C)}) / (1 - C \cdot e^{-N(1-C)})$$

Counterflow, Cr = 0.985; NTU from laminar coil flow with Dean enhancement, 2.4 m of 3/16″ in 3/8″

ε = 0.949

At 1.2 L/h; 0.934 at 1.6 L/h.

2 Heater budget

$$P = \dot{m} c_p (1-\varepsilon) \Delta T + \dot{m}\, f\, h_{fg} + UA(T_b - T_a)$$

5.3 W sensible + 11.3 W steam (1.5 %) + 6.0 W losses
without the exchanger: 121 W

22.5 W

A 60 W heater has headroom while running.

3 Time at the boil

$$t = V_{boil} / Q$$

61 mL ÷ (1.2 L/h = 20 mL/min)

3.05 min

Yu et al. boiled for minutes: 3 min is the minimum we set.

4 Lime formed

$$\text{lime} = H - 20.3 \text{ mg/L}$$

PHREEQC, vented boil at 100 °C
285 − 20.3 mg/L as $CaCO_3$

265 mg/L

0.13 g of lime per 0.5 L jug into the drawer.

5 Microplastic removal

$$R = 1 - e^{-0.00864 \cdot \text{lime}}$$

Fitted to Yu et al. 2024 (25 % soft, 34 % at 80, 84 % at 180, 90 % at 300 mg/L)
1 − exp(−2.29) = 0.90

90 %

At 80 mg/L: 40 %. Hard water is the whole mechanism.

6 Outlet temperature

$$T_{hot} = T_j + (1 - \varepsilon)(T_b - T_j)$$

25 + 0.051 × 74.5 = 28.8 °C leaving the exchanger body coil in the finned shell trims it

27.1 °C

Steady state only; the real run starts with a hot jug.

7 Energy per litre (real run)

$$E/V = \int P\, dt / V_{out}$$

as drawn: 65 Wh ÷ 0.49 L
with pump thermostat: 36 Wh ÷ 0.49 L

133 Wh/L

73 Wh/L with the fix. A 90 Wh bank gives 0.7–1.2 L, not 3 L.

8 What limits flow

$$Q \le V_{boil} / 3 \text{ min} \;;\; P(Q) \le 60 \text{ W}$$

dwell: 61 mL ÷ 3 min = 1.22 L/h
heater: P = 60 W at 2.9 L/h

1.2 L/h max

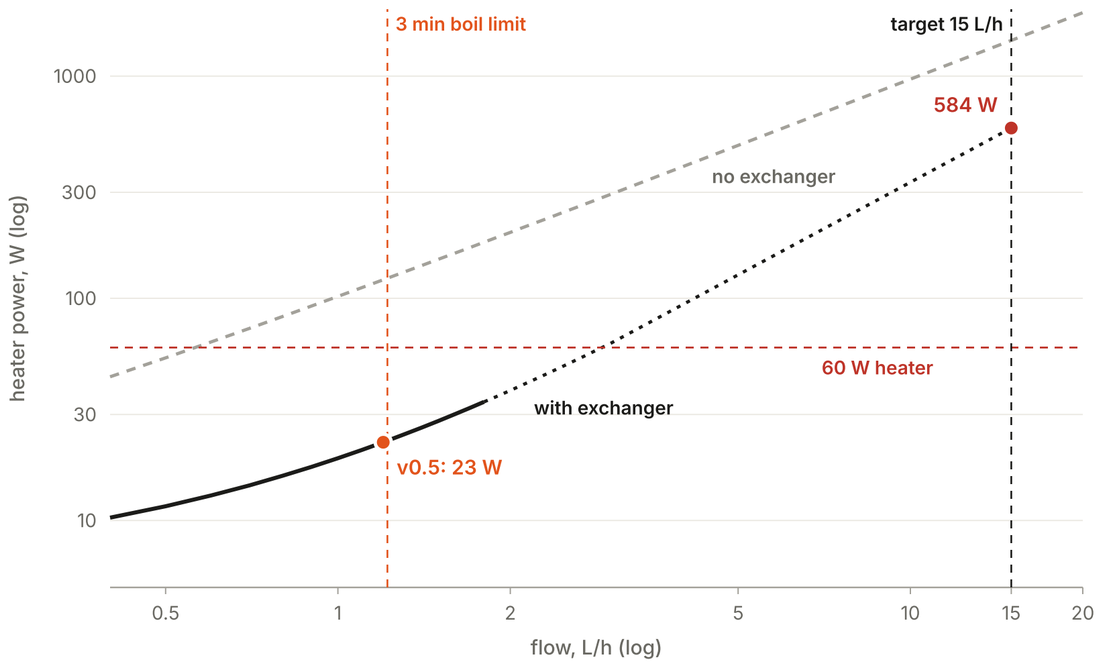
15 L/h needs ~580 W and a 750 mL boil coil.

Inputs: v0.5 drawings (tube sizes, 61 mL boil coil), 60 W heater, 1.5 % steam, loss UA 0.08 W/K, PHREEQC lime, Yu et al. 2024

Result 1 · Flow: 1.2 L/h, a twelfth of our 15 L/h target

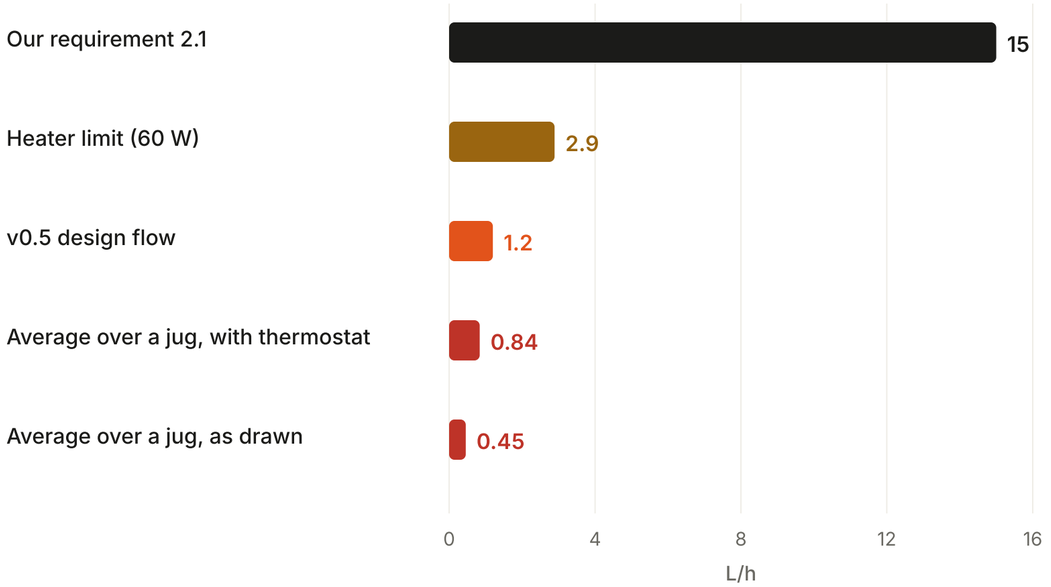
Two limits stack up: the 3 minutes at the boil caps flow at 1.22 L/h, and the 60 W heater caps it at 2.9 L/h. Reaching 15 L/h is a different product.

Heater power needed vs flow (25 °C water)



- The exchanger cuts the heater from 121 W to **23 W** at 1.2 L/h. Above about 1.8 L/h the curve is extrapolated (dotted).
- At 15 L/h even with the exchanger: about **584 W**, far beyond what USB-C can supply (240 W at most).

Litres per hour, compared



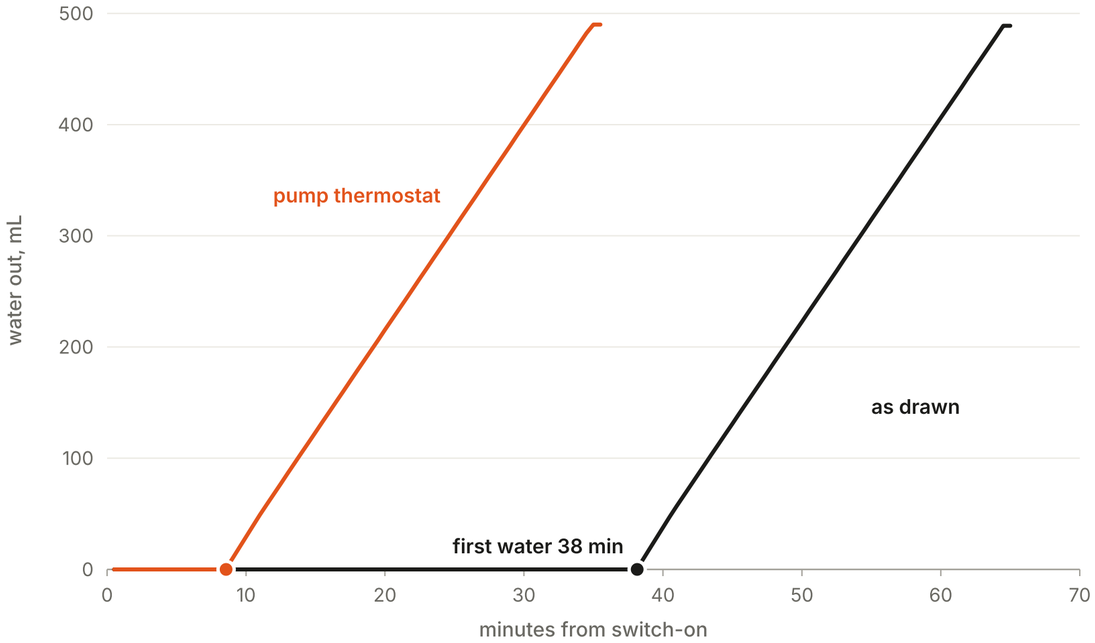
- Counting warm-up, a 0.5 L jug takes **64 min** as drawn (0.45 L/h) or **35 min** with the thermostat.
- 3 L a day (requirement 1.1) means about 2.5 h of running and six jug refills.

model.mjs steady() and run(); ε fit valid to 1.6 L/h, extrapolated beyond

Result 2 · Warm-up decides the first 40 minutes

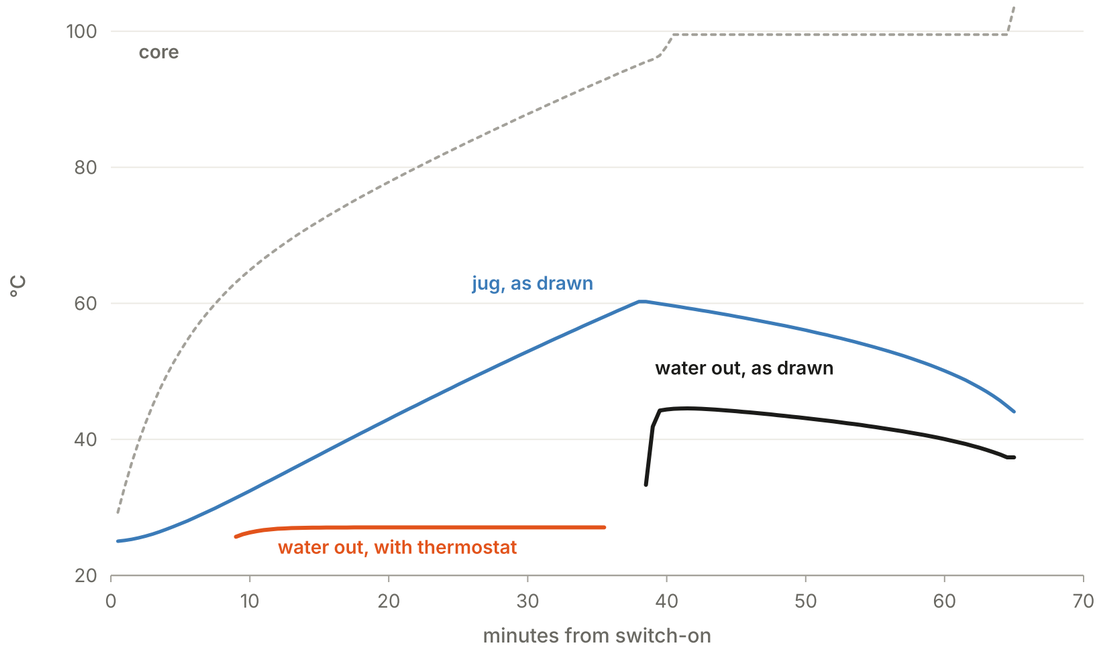
As drawn, the warm-up overflow returns boiling water to the jug. That delays the first water and makes it come out hot. Starting the pump only when the core is ready fixes both.

Water delivered vs time from switch-on



- As drawn: **38 min** to first water (the concept sheet said about 15), jug finished at **64 min**.
- With the thermostat: **9 min** to first water, jug finished at **35 min**.

Temperatures during the run

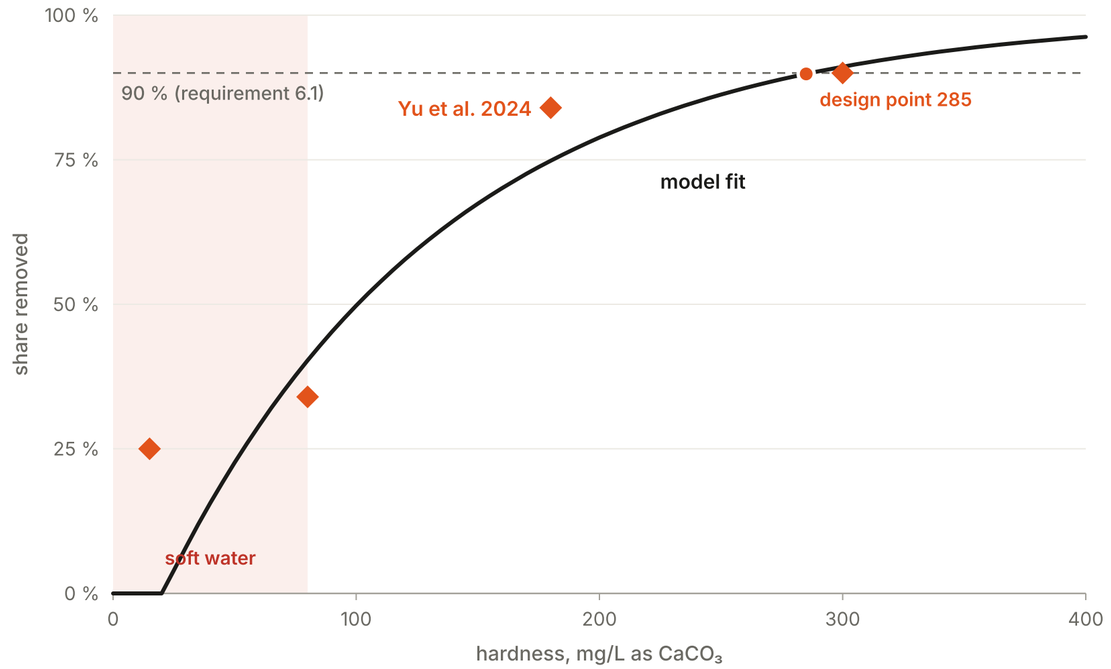


- As drawn, the overflow heats the jug to about 58 °C, so water leaves at up to **45 °C**: drinkable, but not the 27 °C promised.
- With the thermostat the jug stays cold and the outlet holds **27 °C** in a 25 °C room.

Result 3 · Removal depends on the water, energy on the warm-up

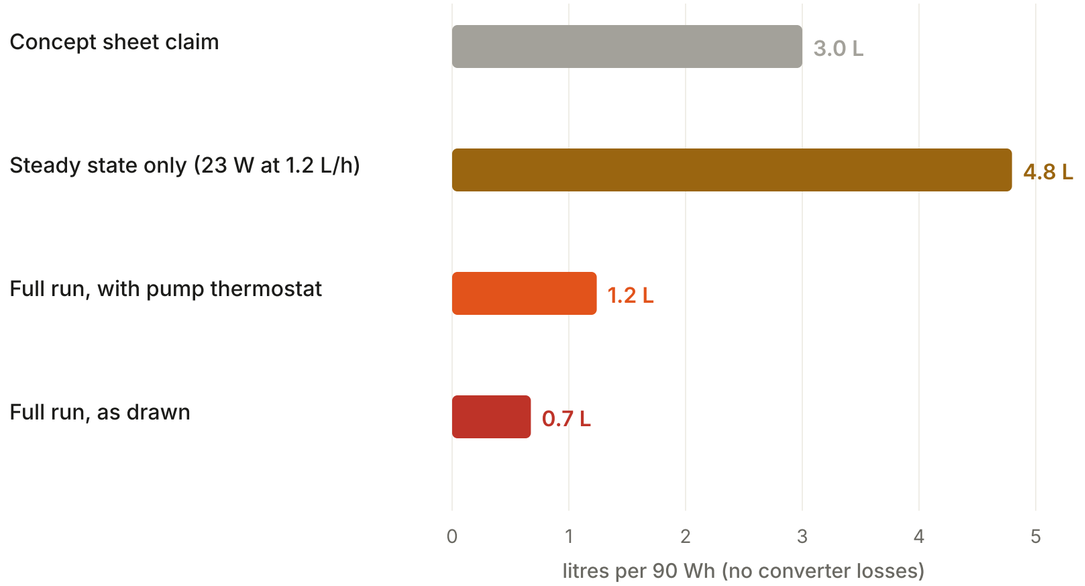
Boiling doesn't remove plastic by itself: lime from hard water traps it. And a power bank goes much less far than the concept sheet said.

Microplastic removal vs water hardness



- Reaches **90 %** only above about **290 mg/L**. At 180 mg/L the fit gives 75 % (Yu measured 84 %); at 80 mg/L about **40 %**.
- So the first test is the hardness of the water we target. The fit stays within about 10 points of Yu's four measurements.

Litres from a 90 Wh power bank



- Warm-up costs more than running: **65 Wh** for one jug as drawn, **36 Wh** with the thermostat.
- So the power bank use case works best for long runs, not single jugs. Mains USB-C is the realistic supply.

Removal: 1 − e^(−0.00864 · (H − 20.3)), fitted to Yu et al. 2024 (Environ. Sci. Technol. Lett. 11, 273–279)

Our 22 technical requirements: what Boil-Loop meets and what it doesn't

Ranked by our weighted objective tree. **Pass · sim**: the simulation shows it. **By design**: the choice should meet it, but nothing has tested it yet.

Pass · sim 1 | Met by design 1 | By design · untested 3 | Partial 4 | Fail 4 | At risk 4 | Not yet tested 3 | Not applicable 2

Rank	#	Requirement	Boil-Loop v0.5 result	Checked by	Status
1	**6.1**	Removal ≥ 90 % by count, 1 µm–5 mm (stretch 99 %)	90 % at 285 mg/L; about 40 % at 80 mg/L; 25 µm mesh catches the crust	Sim	Partial
2	**3.2**	Filter medium sheds no secondary micro/nanoplastic	Steel mesh and tubes; but the PharMed pump tube wears in the water path	Design	At risk
3	**6.2**	Detection false +/− ≤ 10 % vs reference microscopy	No detection in this concept; would pair with C05	—	Not applicable
4	**10.2**	Captured microplastics stay contained	Lime crust in the drawer; rinsing it into a sink releases it	Design	At risk
5	**4.1**	Result through ≥ 1 user feedback channel	Dial thermometer only; no microplastic information	Design	Fail
6	**11.1**	Retail price ₹1,500–6,000	v0.5 not costed; v0.4 BOM implied ₹6,650 retail, v0.5 adds a pump	BOM	At risk
7	**3.1**	Water-contact parts food-contact safe	316L, borosilicate, FDA silicone, PharMed tube	Spec	By design · untested
7	**3.4**	No leachables above IS 10500	316L at 100 °C plus citric descale; not tested	Spec	By design · untested
9	**2.1**	Flow rate ≥ 15 L/h	1.2 L/h (0.45–0.84 L/h including warm-up)	Sim	Fail
9	**1.1**	Per-use capacity for 1–3 L/day	3 L takes about 2.5 h and six refills of the 0.5 L jug	Sim	Partial
9	**10.1**	Disposal pathway for spent filters (W)	Steel mesh is reusable; lime + plastic catch has no defined route	—	Not yet tested
9	**1.2**	Non-destructive sampling (W)	Drawer catch can be sampled, but the plastic is locked in lime	Design	Partial
9	**5.3**	Display legible outdoors (W)	No display; dial thermometer	—	Not applicable
14	**9.1**	Filter rated ≥ 100 L or 3 months	Mesh reusable; cartridge clogging and tube scaling not modelled	—	Not yet tested
14	**7.1**	Filter swap tool-free in ≤ 30 s	Twist-out drawer; not timed	Design	By design · untested
14	**4.2**	Alert near filter end-of-life	None	Design	Fail
14	**5.2**	Spout parts detachable and washable	Spout is a fixed steel tube	Design	Fail
14	**8.2**	Usable at 5–45 °C	Steady outlet = room + 2 °C; shell stays near room temperature	Sim	Pass · sim
19	**3.3**	Housing UV and impact resistant (W)	Aluminium shell; glass jug on top	Design	Partial
19	**3.5**	Survives 1 m drop (W)	2.7 kg with a glass jug; not tested	—	At risk
21	**5.1**	Actuation force ≤ 50 N (W)	No manual actuation	Design	Met by design
21	**8.1**	Noise ≤ 60 dB(A) (W)	Peristaltic pump + boiling; not measured	—	Not yet tested

Only one Tier 1 item is even partly met, and only in hard water. Flow (2.1) misses by 12×. Detection (6.2, 4.1) is outside this concept.

Requirements: MPx_Weighted_Objective_Tree.docx (22 items, D/W). W = wish

What the simulation proves, and what only a bench can

The heat side is well modelled and mostly passes. The chemistry, the scaling and the shedding are assumptions until we boil real water.

ENGINEERING CHECKS FROM THE MODELS

Check	Result	Status
Heater at 1.2 L/h	22.5 W of 60 W	pass
Steady outlet, 25 °C room	27.1 °C	pass
Shell surface	26.8 °C, safe to hold	pass
Time at the boil	3.05 min ≥ 3	pass
Unboiled water blocked	wax gate shut below 92 °C	pass
Removal at 285 mg/L	90 %	pass
Removal at 80 mg/L	40 %	fail
Flow	1.2 L/h vs 15	fail
First water, as drawn	38 min vs ~15 claimed	fail
Outlet in a real run, as drawn	45 °C	fail
90 Wh power bank	0.7 L vs 3 L claimed	fail
Mass	2.73 kg vs 1.3 kg claimed	fail

ASSUMPTIONS TO REPLACE WITH MEASUREMENTS

Assumption	Value used	Drives	Bench test
Lime formed in a vented boil	H − 20.3 mg/L (PHREEQC)	Removal (6.1)	1 · hardness before and after, our tap and borewell water
Removal vs lime	Yu et al. fit, kettle + mesh	Removal (6.1)	2 · PS bead or Nile red count, boiled vs not
3 min is enough	dwell = 3.05 min	Removal, flow	3 · run 2, 4 and 6 min
Exchanger ε with Dean flow	0.949 at 1.2 L/h	Heater, outlet	4 · thermocouples on all four ports
Steam 1.5 %, loss 0.08 W/K	11 W + 6 W	Heater, energy	4 · power meter on USB-C
Tube and mesh scaling	not modelled	Life (9.1), flow	5 · 50 L run, weigh scale, pressure
Pump tube shedding	not modelled	Shedding (3.2)	6 · blank run, count particles out

CED LECTURE 9: THREE KINDS OF SIMULATION

- **Analytical (slide 7)**
- **Virtual (slides 5–6)**
- **Physical: tests 1–6, next**

tube_sizing_sim.py (CoolProp properties), model.mjs (transient), PHREEQC chemistry from the v0.2 study

Design FMEA: method

Pahl & Beitz §10.4: list how each part and use step can fail, what the user notices, and why. Then rate, rank and fix.

1 Risk analysis

Failures, consequences, causes; measures that avoid or detect them

2 Risk assessment

Rate occurrence O, severity S, and detection D (easy to detect = low D)

3 Risk number

RN = O × S × D, from 1 to 1000. **RN > 125 is critical**

4 Risk minimisation

Design change for each critical item, then re-rate for a new RN

Occurrence O

1–2	Unlikely: a design feature prevents it
3–4	Occasional
5–6	Likely over a year of use
7–8	Frequent: most users, often
9–10	Certain: the simulation shows it every time

Severity S

1–2	User barely notices
3–4	Slower or less convenient
5–6	Function degraded
7–8	Hygiene risk, scald or contaminant released
9–10	Untreated water delivered as treated

Detection D

1–2	Obvious at once
3–4	User notices within a use
5–6	Found only by inspection or test
7–8	Hidden from the user
9–10	Cannot be detected

WORKED EXAMPLE: WATER CHEMISTRY

Failure	Effect	Cause	Rating	RN	Action	New RN
Removal collapses in soft water	User thinks the water is clean	Soft target water	O 7 × S 8 × D 8	448 · critical	Hardness strip + removal estimate	4 × 8 × 3 = 96

Scope: v0.5 as drawn (jug, pump, exchanger, heater, boil coil, lime drawer, wax gate, body coil, electrics) and three use steps (warm-up, producing, cleaning). **20 failure modes, 11 critical.** Ratings use the model where it gives numbers; the rest is judgement for the team to review.

Pahl & Beitz, Engineering Design 3e (2007), §10.4 and Fig. 10.11; CED lecture 2, slide 36

Design FMEA (1/2): the ten highest risks

Sorted by risk number. Red RN is critical (> 125). New RN is the re-rating once the action is in the design.

#	Part / function	Failure mode	Effect on user	Cause	Current control	O	S	D	RN	Recommended action	New RN	Req.
1	**Water chemistry**	Removal collapses in soft water	User thinks the water is clean; 25–40 % removed	Target water below ~150 mg/L hardness	Note on the concept sheet	7	8	8	448	Hardness strip in the box; show expected removal per hardness; test a calcium-dosing cartridge	96	6.1
2	**Drawer rinse**	Caught crust rinsed into the sink	Microplastics go back to the environment	Twist-out drawer is rinsed under a tap	None	8	6	8	384	Dissolve the crust in citric acid in a cup, filter on paper, dry and bin	72	10.2
3	**Pump tube (PharMed)**	Tube wear sheds particles	The purifier adds polymer particles	Rollers abrade the tube every turn; spallation	Pump on the cold side	6	7	8	336	Steel diaphragm or gear pump, or gravity feed (v0.4); blank-water shedding test	105	3.2
4	**Mini tubes and boil coil**	Lime scales the walls	Flow drops, heater overheats, pump stalls	The same lime that traps plastic deposits on hot walls	None	8	6	6	288	Automatic citric-acid descale cycle; smooth 316L; pressure check on the pump	96	9.1
5	**Boil dwell**	Too little time at the boil	Less lime forms, removal falls	Flow set high; scale shrinks coil volume	Fixed pump speed	5	7	7	245	Lock flow at ≤ 1.2 L/h; dwell test at 2/4/6 min	105	6.1
6	**Wax gate**	Sticks open	Unboiled water reaches the spout	Wax actuator failure; lime on the seat	None	3	8	8	192	Gate after the mesh, away from lime; cycle test; dial check in the manual	80	6.1
7	**Tube joints (30+ fittings)**	Hot leak	Hot water in the base near electrics	Thermal cycling of compression fittings	20 kPa leak test	5	7	5	175	Brazed joints in production; drip path away from electrics	56	3.5
8	**Tubes between runs**	Stagnant water grows biofilm	Odour; first water from the cold side not boiled	Tubes stay primed between uses	Next run boils it	5	5	6	150	Drain mode; first 60 mL to waste	45	3.1
9	**316L at 100 °C**	Metal leaching after descale	Ni/Cr above IS 10500	Acid descale, weld tint, poor passivation	316L	3	8	6	144	Passivate after descale; leach test after 20 cycles	32	3.4
10	**Lime-trap cartridge**	Mesh clogs	Back-pressure, overflow, no output	~0.13 g of lime per jug at 285 mg/L	Twist-out rinse	7	5	4	140	Size the cartridge for a week; overflow is the visible cue	60	9.1

Design FMEA (2/2): the remaining ten

One more critical item: steam bursts. The flow row shows where RN falls short: a certain, obvious failure scores low.

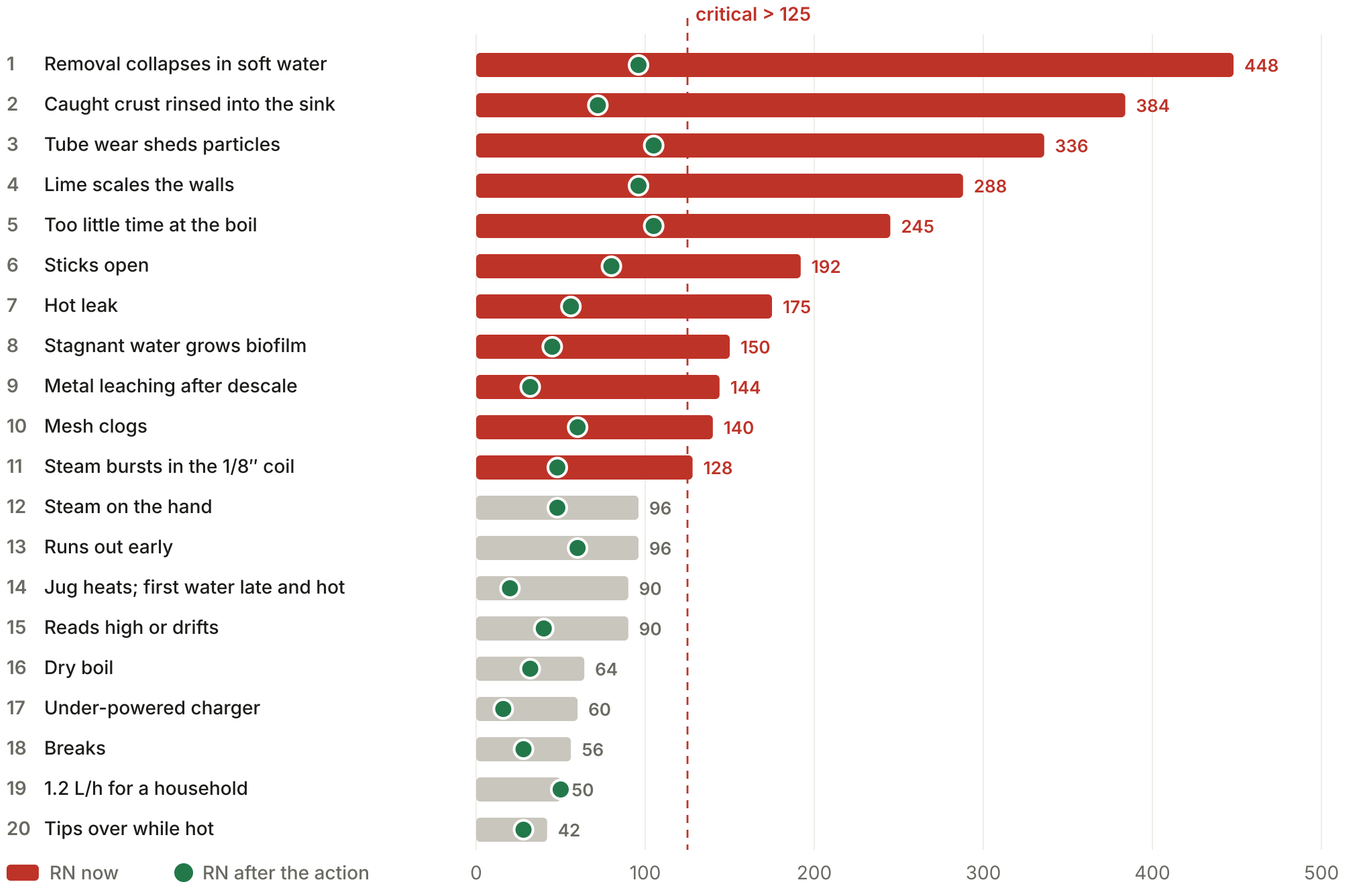
#	Part / function	Failure mode	Effect on user	Cause	Current control	O	S	D	RN	Recommended action	New RN	Req.
11	**Mini-tube heater**	Steam bursts in the 1/8″ coil	Hot water spurts from the vent or spout	Boiling in a 3 mm tube	Boil only in the vented coil	4	8	4	128	Cap heater-coil outlet at ~98 °C; vent splash guard	48	—
12	**Vent and cap**	Steam on the hand	Scald when lifting the jug	Vent standpipe exits at the top	Insulated core	4	8	3	96	Vent shield; cool-touch cap	48	—
13	**Power bank use**	Runs out early	Only ~0.7 L per 90 Wh	Warm-up energy	None	8	4	3	96	Pump thermostat (~1.2 L); market as mains USB-C	60	1.1
14	**Warm-up overflow**	Jug heats; first water late and hot	38 min wait; 45 °C out	Overflow returns ~100 °C water to the jug	None	9	5	2	90	KSD301 98 °C pump thermostat: 9 min, 27 °C	20	8.2
15	**Dial thermometer**	Reads high or drifts	User trusts unboiled water	Bimetal drift	None	3	5	6	90	Calibrate at the boil; gate is the real safeguard	40	4.1
16	**Heater, jug empty**	Dry boil	Smell; scale baked on	Pump keeps running dry	130 °C KSD301 + 152 °C TCO	8	4	2	64	Pump stops with the heater cut-off	32	—
17	**USB-C supply**	Under-powered charger	Never reaches the boil; no water	45 W charger or PD at 15 V	20 V PD trigger	5	4	3	60	LED for 20 V; 65 W on the box	16	—
18	**Glass jug**	Breaks	Cuts, spill	Drop, knock	Borosilicate 3.3	4	7	2	56	Steel jug option with a sight window	28	3.5
19	**Flow**	1.2 L/h for a household	Too slow for daily use	3 min dwell, 60 W heater	None	10	5	1	50	Not fixable within the concept; reposition as a slow batch purifier	50	2.1
20	**Whole unit**	Tips over while hot	Hot spill	388 mm tall on a 170 × 120 base	None	3	7	2	42	Wider base; tilt switch on the heater	28	3.5

Why flow scores only 50: it happens every time (O 10) but everyone notices at once (D 1), so the RN is low. FMEA ranks hidden risks; it doesn't judge whether the concept meets its requirements. That is what slide 11 does.

FMEA result: 11 critical risks, mostly chemistry and hygiene

The heat side is in good shape. The risks are in what the water does to the device (scale) and what the device could add to the water (pump tube, leaching).

Risk number per failure mode, before → after



CHANGES FOR A V0.6, IN PRIORITY ORDER

1. **Hardness gate**
   Strip test in the box and a removal estimate per hardness (6.1)
2. **Crust disposal**
   Citric-acid dissolve, paper filter, dry and bin (10.2)
3. **Pump without a wearing tube**
   Steel diaphragm or gear pump, or gravity feed as in v0.4 (3.2)
4. **Automatic descale cycle**
   Citric acid through the coils on a schedule (9.1)
5. **Pump thermostat**
   KSD301 98 °C: first water in 9 min at 27 °C
6. **Flow lock and gate placement**
   ≤ 1.2 L/h; wax gate after the mesh
7. **Brazed joints, drain mode**
   Fewer leak paths; no stagnant water
8. **Steam shield, wider base, 65 W label**

None of these fixes the flow. After the actions every RN is at or below 105, but 1.2 L/h remains. That is a product decision, not an FMEA one.

Next steps: two cheap tests decide whether to continue

The heat side is proven on paper. Before spending on v0.6, boil our own water and decide whether 1.2 L/h is acceptable.

WEEK 1

Hardness of our water

- Strip or EDTA titration: IISc tap, borewell, packaged
- If < ~200 mg/L: removal < 80 %

WEEK 1

Decide on flow

- Is 1.2 L/h acceptable as a slow, overnight purifier?
- If not: stop or rethink

WEEK 2

Kettle bench test

- Boil 2/4/6 min, filter on 25 µm mesh
- Count with Nile red, boiled vs not

WEEK 2–3

Heat-side rig

- Tube-in-tube + heater on a bench PSU
- Measure ε, power, warm-up

WEEK 4

Close the loop

- Measured values into model.mjs
- Re-run requirements and FMEA

THEN

v0.6 or stop

- v0.6 with the FMEA changes
- or keep as a lab method for C05 samples

BENCH TESTS: WHAT EACH ONE MEASURES, AND THE REQUIREMENT IT SETTLES

1. **Hardness before and after boiling**
   Measures lime actually formed
   settles 6.1
2. **Nile red count, boiled vs unboiled**
   Measures real removal on our water
   settles 6.1, 1.2
3. **Boil for 2, 4 and 6 min**
   Measures minimum time at the boil
   settles 6.1, 2.1
4. **Thermocouples + USB-C power meter**
   Measures ε, heater power, warm-up
   settles 2.1, 1.1
5. **50 L endurance run**
   Measures scale on tubes and mesh, pressure
   settles 9.1
6. **Blank-water run through the pump**
   Measures particles shed by the pump tube
   settles 3.2
7. **Leach test after 20 descales**
   Measures metals vs IS 10500
   settles 3.4
8. **Drop, tip and leak checks**
   Measures joints, jug, stability
   settles 3.5

Tests 1–3 need only a kettle, a mesh and the Nile red kit already planned for C05